

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Zen Black (v2.3 SP1 FP3) , Leica Application Suite X (v3.7.4.23463; v3.8.1.26810; v4.7.0.28176), Sony Cell Sorter Software (v2.1.6), BD FACSDiva (v 8.0.1), CytExpert SRT (Version 1.0.3.10011)
Data analysis	Imaris (v8.0 - 10.0); R (v4.3.1); R Studio (2023.090+463); Python (v3.10.4); Seurat (v4.4.0); scVI (v1.0.4) ; inferCNV (v 1.18.1); cNMF (v1.4.1); pySCENIC (v0.12.1); 10x Cellranger (v3.1); DESeq2 (v1.40.2); VoxHunt (v1.0.1); gprofiler2 (v0.2.2) ;FlowJo Software (v10.9.0); GraphPad Prism (v.8.0.2); Microsoft Excel (Version 2312 Build 16.0.17126.20190); QuPath (v0.4.4); SCEVAN (v 1.0.1); scclusteval (v.1.0); ImageJ2 (v 2.3.0/1.53q); xPONENT (version 4.3u1); Bio-Plex Manager (version 6.2). All used R and Python scripts for analysis, excluding lineage tracing, are available at the Rios' Dream3D laboratory's GitHub page, https://github.com/Dream3DLab/DMGO_analysis . Pipelines for analyzing TrackerSeq data can be found at the Alemany's group GitHub, https://github.com/anna-alemany/TrackerSeq_BROs .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Raw sequencing and methylation data that support the findings of this study have been deposited in the European Genome Phenome Archive (<https://ega-archive.org/>) under accession code E-MTAB-15147 and E-MTAB-15559, respectively. Processed sequencing data have been deposited in Zenodo (<https://zenodo.org>) under accession code 10.5281/zenodo.16992353. Sequencing metadata is provided in Supplementary Table S2, S3, S4, S6 and S9.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Sex was not considered during study design.
Reporting on race, ethnicity, or other socially relevant groupings	No categorization variables were included in this study.
Population characteristics	For this manuscript only patients with DMG tumours harboring H3.3K27M, TP53 and PDGFRA mutations were included.
Recruitment	Samples were obtained as part of an institute wide bio-banking effort. All brain tumour patients and/or parents or guardians were approached for participation, which entailed inclusion of remaining biopsy/ autopsy material.
Ethics oversight	For the use of samples from patients with DMG, patients and/or parents or guardians provided written informed consent according to national laws and in agreement with the declaration of Helsinki (2013). This study is Institutional Review Board (IVB) approved and registered under national registry number 2020.142.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Organoids from well established cell lines (e.g. WA09/01) show variability from batches and not individual organoids (see ref79). Hence, a consistent phenotype was considered significant and consistent when shown in minimum of three batches/biological replicates. Further we considered the same significance and consistency of biological replicates for tumor induction and heterogeneity.
Data exclusions	An individual organoid not matching correct differentiation (control performed with markers: PAX7, MYOG, MYOD1; occasionally can occur due to few similarities in developmental cue's relevant for mesodermal fate e.g. CHIR99021, see Lullo and Kriegstein 2017) and subsequent tumor induction was excluded. Additionally samples with low cell number (below 100 cells) counts after filtering were excluded. Further one sample was excluded from large versus small barcoded due to non-clonal behaviour (a single dominant barcoded clone) making Differential Expressed Gene analysis impossible. All excluded samples and criteria are reported accordingly see Supplementary Table S4, S6 & Methods.
Replication	All quantifications were done on minimum of three independent batches including at least three organoids. All attempts of replications were successful. For accurate listing of batches and organoids used in this study, see Supplementary Table S1, detailing batch versions, cell source, protocol applied etc.
Randomization	Organoids as well as tumor-bearing organoids were cultivated and subsequently picked randomly from multiple researchers involved in the study for any experiments involving sequencing and imaging.
Blinding	No techniques of blinding the investigators were applied in this study, as there were no case-controls relevant.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

As listed in Supplementary Table S10:

Primary:

SOX2 1:200 Thermo Fisher, 14-9811-82

HOXB1 1:200 R&D Systems, AF6318

TUBB3 1:2500 R&D Systems, MAP1195

TPH2 1:200 GeneTex, GTX101962

NF200 1:500 Sigma-Aldrich, N5389

GFAP 1:1000 Abcam, ab4674

AQP4 1:200 Sigma-Aldrich, MABN2527

OLIG2 1:200 Sigma-Aldrich, AB9610

GFP 1:200 Abcam, AB6673

GFP 1:200, Abcam, AB13970

p53 1:100 Invitrogen, 13-4100

PDGFRA 1:250 Cell Signaling, 8871

H3K27M 1:100 Invitrogen, MA5-27916

H3K27M (H3K28M) 1:400 Abcam, AB190631

H3K27me3 (H3K28Me3) 1:200 Cell Signaling, 97335

GFAP RTU Leica, PA0026

NF RTU Leica, PA0371

Iba1 1:50, Fujifilm Wako, 019-19741

P2RY12 1:50 Atlas, HPA014518

CD163 1:50 Invitrogen, MA5-11458

SPP1 1:50 R&D Systems, AF1433-SP

CD3 1:100 BioLegend, 300402

Iba1 1:100 Novus Biologicals, NB100-1028SS

cCasp3 1:100 Cell Signaling, 9661S

CD31 1:400 Abcam, AB134168

Secondary:

anti-rat AF594 1:250 Invitrogen, A21209

anti-sheep AF555 1:250 Invitrogen, A21436

anti-mouse IgG2a AF488 1:250 Invitrogen, A21131

anti-rabbit AF647 1:250 Invitrogen, A31573

anti-mouse IgG1 AF568 1:250 Invitrogen, A21124

anti-chicken AF647 1:250 Invitrogen, A21449

anti-mouse AF594 1:250 Invitrogen, A11032

anti-goat AF488 1:250 Invitrogen, A11055

anti-chicken AF488 1:250, Invitrogen, A11039

anti-rabbit AF594 1:250, Invitrogen, A21207

anti-mouse AF555 1:250, Invitrogen, A31570

Bond Polymer Refine Detection - Leica, DS9800

anti-rabbit Cy5 1:200, Jackson ImmunoResearch, 711-175-152

anti-mouse AF488 1:200, Invitrogen, A21202

anti-goat AF555 1:200, Invitrogen, A32816

anti-mouse AF555 1:250, Invitrogen, A32773

anti-goat AF647 1:250, Invitrogen, A21447

anti-rabbit AF488 1:200, ThermoFisher, A21206

Conjugated:

CD3-APC 1:80 BioLegend, 344811

NCAM1-HiLyte 488 1:200 QVQ, FSH-10B10
 GD2-PE 1:200 BD Biosciences, 562100
 CD3-BV421 1:100 BD Biosciences, 563798
 GFP-booster ATTO647N 1:400, ChromoTek, GBA647N-100

Validation

Individual antibodies were commercially available and thus validated based on published data as well as manufacturers information and validation procedures. Further signal-location and signal-to-noise levels were critically evaluated based on available information. We refer to the individual manufacturers procedures to ensure integrity of the offered antibodies, e.g. <https://www.sigmaaldrich.com/NL/en/products/protein-biology/antibodies>.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

H9 (WA09, Wicell; sex: Female), H1 (WA01, Wicell; sex: Male) and iPSC C7-a (RUID 06C52463, RUCDR Infinite Biologics; sex: Female)

Authentication

H9 (WA09, Wicell) and H1 (WA01, Wicell) were authenticated by WiCell via STR profiling. iPSC C7-a (RUID 06C52463, RUCDR Infinite Biologics) provided by Marius Wernig lab was not authenticated separately.

Mycoplasma contamination

All cells were tested for mycoplasma and tested negative.

Commonly misidentified lines
(See [ICLAC](#) register)

No commonly misidentified cell lines were used.

Animals and other research organisms

Policy information about [studies involving animals; ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ (NSG) mice (The Jackson Laboratory, #005557) were housed at 45-65% humidity, 20.5-23.5 °C and 12-hour light/12-hour dark cycle, in Specific Opportunist Pathogen Free (SOPF) conditions using Individually Ventilated Cages (IVC) and received sterile food and water ad libitum.

Wild animals

No wild animals were used in this study.

Reporting on sex

Both male and female mice were used for this study. Sex was not considered during study design.

Field-collected samples

No field-collected samples were used in this study.

Ethics oversight

All murine experiments were conducted in compliance with the Animal Welfare Body of the Princess Máxima Center for Pediatric Oncology based on local and international regulations under CCD license AVD39900 202216507.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

DMG organoids were dissociated into single cells using the Neural Tissue Dissociation Kit (P) (Miltenyi Biotec, Cat. #130-092-628). After cutting the organoids into smaller pieces and adding the enzyme mixes according to the manufacturers instructions, the samples were incubated on an orbital shaker at 37°C and resuspended in regular intervals with a P1000 until a single cell suspension was reached, which was verified under the microscope using trypan blue to check for cell death. After dissociation, the single cell suspension was washed twice with PBS +/- (magnesium/calcium+ 3% FBS). Each organoid was then stained with a different TotalSeq-A anti-human hashtag (Biolegend Cat #394601, #A0251-A0265), allowing us to tag each cell with a sample-specific artificial oligonucleotide that can be recovered by sequencing. The hashing oligos were designed to recognize most human cells using a combination of two clones against CD298 and β 2 microglobulin. The cell pellet was resuspended in staining buffer (50 μ l for 500,000 cells). Unspecific binding was reduced by adding 5 μ l of human Fc blocking reagent to the sample (FcX Human truStain, Biolegend #422301). After 10 min incubation at 4 °C, 1 μ l of a unique cell hashing antibody was added to each sample and incubated for 20 min at 4 °C and then washed 3x using PBS +0.04% of BSA. Next, DAPI was used as a viability dye (Dapi 1:5000). Stained single cells suspensions were filtered using a 40 μ m Flowmi cell strainer and sorted by FACS on DAPI exclusion and enriched for GFP expression for tumour cells. Organoids treated with T cells were instead of DAPI stained with LIVE/DEAD Fixable Near-IR Dead Cell Stain (1:1000; ThermoFisher, L34976) and CD3-APC (1:80; BioLegend, 344811) for 30 minutes at 4 °C. Stained single cells suspensions were filtered using a 40 μ m Flowmi cell strainer and sorted by FACS on LIVE/DEAD dye exclusion and enriched for GFP expression for tumour cells or CD3 for T cells. FACS experiments were performed using the CytoFLEX SRT Benchtop Cell Sorter (Beckman Coulter).

To confirm GD2 expression of the organoids, cells were retrieved from the culture plate using StemPro Accutase (Gibco, Cat. #A1110501) and passed through a 70 μ m Flowmi cell strainer (Merck, Cat.#BAH136800070) to create a single cell suspension. Dissociated cells were centrifuged at 500 RCF for 5 minutes at 4 °C and resuspended and washed in FC buffer (2% fetal bovine serum (FBS), 1x PBS). Cells were either left unstained, or stained with LIVE/DEAD Fixable Near-IR Dead Cell Stain (1:1000; Thermo Fisher, L34976) and GD2-PE (1:200; BD Biosciences, Cat.#562100) for 30 min at 4 °C. After staining, cells were washed twice in FC buffer, acquired on a Sony SH800s (Sony Biotechnology).

For selection of GD2 CAR-T cells on NCAM1 expression, cells were washed and stained for 30 min at 4 °C in FC buffer with LIVE/DEAD Fixable Near-IR Dead Cell Stain (1:1000; Thermo Fisher, L34976), CD3-APC (1:80; BioLegend, 344811) and NCAM-1-HiLyte 488 (1:200; QVQ, FSH-10B10). CD3+ NCAM1- or CD3+ NCAM1+ GD2 CAR T cell populations were separated by FACS on a BD FACSAria II Cell Sorter (BD Biosciences).

For scRNAseq of microglia and GD2 CAR-T cells, cells were washed and stained in FC buffer with CD3-BV421 (1:100; BD Biosciences, 563798) and LIVE/DEAD Fixable Near-IR Dead Cell Stain (1:1000; Thermo Fisher, L34976) for 30 min at 4 °C. CD3 + T cells and mScarlet+ microglia/ PMP were sorted into 386-well plates containing well-specific barcoded primers (Single Cell Discoveries), one cell per well, on a Sony SH800s (Sony Biotechnology).

Instrument

CytoFLEX SRT V5-B2-Y5-R3 Benchtop Cell Sorter (15 Colors, 4 Lasers), Product number is C71883 Serial number: BE24057 (Beckman Coulter, Brea, CA, USA)
Sony SH800s (Sony Biotechnology)
BD FACSAria II Cell Sorter (BD Biosciences)

Software

CytExpert SRT (Version 1.0.3.10011), Sony Cell Sorter Software (v2.1.6) and BD FACSDiva (v 8.0.1) was used for collection and FlowJo Software (v10.9.0) for analyzing the data.

Cell population abundance

The sorting of tumor cells was a purity sort based on the highest expressing GFP+ cells. GFP+ cells were ranging from 1-38% dependent on the samples. PE positive cell ranged from 0-26% and CD3-APC positive cells ranged from 0-61% depending on the samples. Purity post sorting was not assessed to avoid losing cells, however purity can be estimated based on plasmid integration from the sequencing results see tumor and healthy classification in the Methods section.
GD2-PE positive tumor cells of the organoid were 90%.
mScarlet+ microglia abundance in BrO and DMGO organoids ranged from 1-7%, CD3-BV421+ cells ranged from 8-39%

Gating strategy

To remove debris cells were gated on FCS-A/SSC-A. Single cells were gated based on FSC area and height. Next live cells were selected based on exclusion of DAPI or LIVE/DEAD Fixable Near-IR Dead Cell Stain positive cells. Finally mScarlet+ microglia, CD3-BV421+ T cells or GFP+ tumor cells were sorted. For some experiments, GFP- cells were further sorted based on CD3-APC and PE.

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.